Activity 1: N-gram with Backoff

Penalty = 0.4

The exact scores depend on the N-gram probabilities provided in your activity sheet. Those probabilities are not visible in the screenshots, so use the following method.

a) "the cat sat"

- **Level used:** Trigram
- **Score:** P (sat | the cat), if the trigram is available.
- If unavailable, back off to the bigram or unigram level and apply the penalty of 0.4 as specified.

b) "dog ate fish"

- **Level used:** Trigram, if available.
- **Score:** P (fish | dog ate), if the trigram is available.
- Otherwise, back off to a lower level with the penalty.

c) "dog sat fish"

- **Level used:** Trigram, if available.
- **Score:** P (fish | dog sat), if the trigram is available.
- Otherwise, back off to a lower level with the penalty.

Note: The exact numerical scores require the probability table or corpus counts.

Activity 2: Deleted Interpolation

Part A (2 marks)

Given:

$$\lambda_1 = 0.1,\ \lambda_2 = 0.3,\ \lambda_3 = 0.6$$

Calculate:

$$P(\text{sat} \mid \text{the cat}) = 0.1P(\text{sat}) + 0.3P(\text{sat} \mid \text{cat}) + 0.6P(\text{sat} \mid \text{the cat})$$

Answer: Substitute the unigram, bigram, and trigram probabilities to obtain the numerical result.

Part B (4 marks)

Held-out sentence: **"the dog ate fish"**

Trigrams:

1. "the dog ate"
2. "dog ate fish"

Set 1:

$$\lambda = (0.1|0.3|0.6)$$

For each trigram:

$$P_1 = 0.1P(\text{word}) + 0.3P(\text{word} \mid \text{previous word}) + 0.6P(\text{word} \mid \text{previous two words})$$

Set 2:

$$\lambda = (0.3|0.6|0.1)$$

$$P_2 = 0.3P(\text{word}) + 0.6P(\text{word} \mid \text{previous word}) + 0.1P(\text{word} \mid \text{previous two words})$$

Calculate the natural log scores:

$$\text{Total score} = \ln(P_{\text{trigram 1}}) + \ln(P_{\text{trigram 2}})$$

Which set is better? Choose the set with the higher total log probability. The exact answer requires the probability values.

Activity 3: Entropy and Perplexity

Part A (2 marks)

Given:

- Sunny: $p = 0.5$
- Cloudy: $p = 0.25$

- Rainy: $p = 0.25$

Entropy formula:

$$H = -\sum p \log_2 p$$

Substitute:

$$H = -[0.5\log_2(0.5) + 0.25\log_2(0.25) + 0.25\log_2(0.25)]$$

Since $\log_2(0.5) = -1$ and $\log_2(0.25) = -2$,

$$H = -[0.5(-1) + 0.25(-2) + 0.25(-2)]$$

$$H = -[-0.5 - 0.5 - 0.5]$$

$$\boxed{H = 1.5 \text{ bits}}$$

Perplexity:

$$PP = 2^H$$

$$PP = 2^{1.5}$$

$$\boxed{PP \approx 2.828}$$

Part B (4 marks)

Question: List the words that come after "the" and their probabilities. Find the entropy and perplexity of guessing the next word after "the". What is the entropy of the next word after "sat"?

The exact answer requires the sentence corpus used in your activity sheet.

Formulas:

$$H(\text{next word} \mid \text{the}) = -\sum_w P(w \mid \text{the}) \log_2 P(w \mid \text{the})$$

$$PP = 2^H$$

For the next word after "sat":

$$H(\text{next word} \mid \text{sat}) = -\sum_w P(w \mid \text{sat}) \log_2 P(w \mid \text{sat})$$

Note: To calculate the exact answers for Activities 1, 2, and 3 Part B, I need the N-gram probability table or the training sentences used in your worksheet.